



OBJECT ORIENTED SOFTWARE ENGINEERING

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V-Model

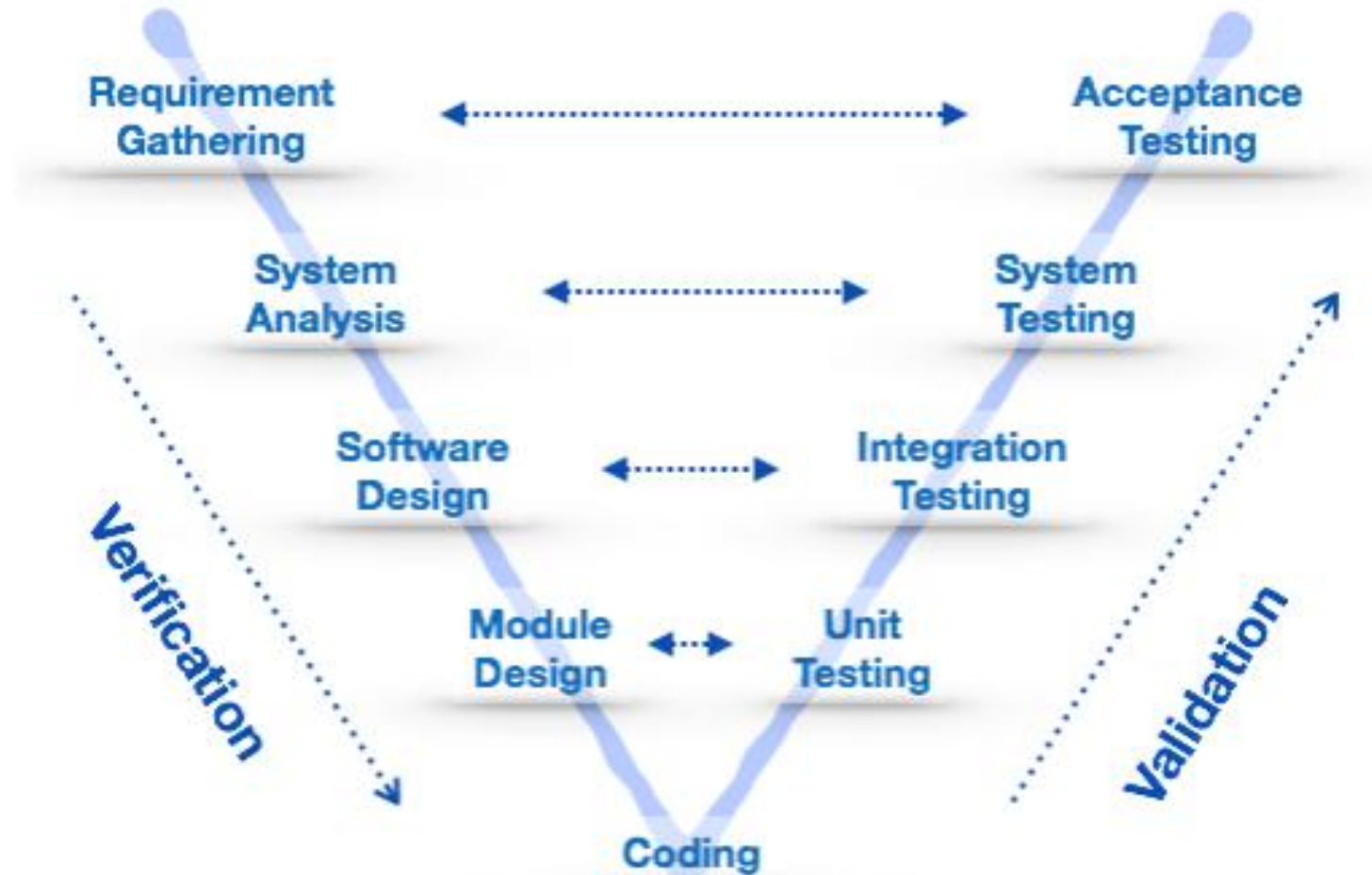
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- There is **no fundamental difference** between the **waterfall model** and the **V-model**.
- The V-model provides a way of **visualizing how verification and validation actions** are applied to earlier engineering work.
- The major drawback of waterfall model is **we move to the next stage only when the previous one is finished** and there was no chance to go back if something is found wrong in later stages. V-Model provides means of testing of software at each stage in reverse manner.

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The V-Model



- A **variation** in the representation of the **waterfall** model is called the ***V-model***.
- At every stage, **test plans** and **test cases** are created to verify and validate the product according to the requirements of that stage.
- For example, in requirements gathering stage the testing team prepares all the test cases in correspondence to the requirements. Later, when the product is developed and is ready for testing, test cases of this stage verify the software against its validity towards requirements at this stage

- This makes both verification and validation go in parallel. This model is also known as **verification and validation model**.
- From the fig, As a software team moves down the left side of the **V**, basic **problem requirements** are refined into progressively more detailed form

- **Once code has been generated** the team moves up the right side of the V, essentially **performing a series of tests** (Quality assurance actions) that validate each of the models created.
- V- model provides a way of visualizing how **verification & validation** actions are applied to **earlier engineering work**.

- This model is used in the **medical development field**, as it is strictly a disciplined domain.
- The following pointers are some of the most suitable scenarios to use the V-Model application.
 - Requirements are well defined, clearly documented and fixed.
 - Product definition is stable.
 - Technology is not dynamic and is well understood by the project team.
 - There are no ambiguous or undefined requirements.
 - The project is short.

The **advantages** of the V-Model method are as follows –

- This is a **highly-disciplined model** and Phases are completed one at a time.
- Works well for smaller projects where requirements are very well understood.
- Simple and easy to understand and use.
- Easy to manage due to the rigidity of the model. Each phase has specific deliverables and a review process.

The disadvantages of the V-Model method are as follows –

- High risk and uncertainty.
- Not a good model for complex and object-oriented projects.
- Poor model for long and ongoing projects.
- Not suitable for the projects where requirements are at a moderate to high risk of changing.
- Once an application is in the testing stage, it is difficult to go back and change a functionality.
- No working software is produced until late during the life cycle.



THANK YOU

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